

Ramy Lazghab

AI & Machine Learning Engineer

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Profile

AI engineer building LLM, agentic AI, RAG and predictive ML systems from prototype to production, with a focus on evaluation and reliability. M.Sc. in Artificial Intelligence & Data Science from Université Paris Dauphine – PSL.

Core Technical Skills

Programming: Python, SQL, Java, JavaScript, C, R

Machine Learning: Deep Learning, Reinforcement Learning, NLP (Natural Language Processing), Transformers, Fine-tuning, Gradient Boosting, Model Explainability (SHAP); PyTorch, TensorFlow, XGBoost, LightGBM

Generative AI & LLM: RAG, Multi-Agent Systems, LLM Evaluation, Prompt Engineering; LangChain, LangGraph, Hugging Face, Anthropic API, OpenAI API, MCP, LangSmith

MLOps, Cloud & Data: Docker, Git, MLflow, GCP (Vertex AI, BigQuery, Cloud Run), REST APIs, PostgreSQL, Neo4j, vector databases (FAISS, Qdrant), Spark, Pandas, NumPy, Streamlit

Professional Experience

AI Engineer Intern

Apr – Oct 2026

Relay X, Paris

- **Oracle AI, regulatory risk forecasting:** predicts whether a regulator will restrict an ingredient within 6 or 18 months. 1,222 scientific opinions encoded as 70 features; XGBoost + LightGBM ensemble, isotonic calibration, SHAP explanations. **AUC-ROC 0.948 at the 6-month horizon and 0.914 at 18 months**, under strict temporal validation.
- Live enrichment through a LangGraph pipeline with three parallel fetch nodes (PubMed, PubChem, web), rendered as a Streamlit intelligence report; LangSmith tracing.
- **PIF (dossier generation):** six-agent LLM pipeline producing EU Product Information Files (Regulation (EC) 1223/2009) from client documents, with template-driven sections and human validation. **Shipped into the production platform** (Next.js, TypeScript, Express, Prisma, PostgreSQL).
- Authored a golden-case benchmark (5 products, 203 ground-truth fields) that drove three pipeline versions: **unsupported claims cut from 87.5% to 35.0%, F1 raised from 81.8 to 89.5, at \$0.13 per dossier**.
- **Magnet (regulatory radar):** five-tier search cascade (SQL first, LLM only on a miss), three-pass source-cited reporting; benchmarked across 35 reports at **88.6% source provenance**; 75 API security tests.

Software Developer Intern

Feb – Jun 2024

Infinity Management

- Built an e-learning platform: Django REST back end, React front end, Docker containerization.
- BERT sentiment analysis on learner feedback to personalize learning paths.
- Designed the REST APIs and data model, then deployed the application.

Education

M.Sc. Artificial Intelligence & Data Science

2024 – 2026

Université Paris Dauphine – PSL

B.Sc. Computer Engineering (IoT)

2021 – 2024

Faculty of Sciences of Tunis

Selected Projects

- **TelecomPlus:** multi-agent customer support (RAG + SQL routing), LLM-as-a-Judge evaluation, Langfuse & MLflow tracking. *LangChain, LangGraph, FAISS, PostgreSQL*
- **NoSQL Query Generation:** plain English into MongoDB, Neo4j, Redis and RDF queries. *Python, MCP, Docker*
- **Movie Recommender:** end-to-end recommendation system on GCP, exposed as a REST API. *BigQuery ML, Vertex AI, Cloud Run*
- **Alzheimer's Prediction:** ensemble of XGBoost and deep learning for early diagnosis; SHAP for clinical interpretability. *PyTorch, XGBoost, SHAP*

Certifications & Achievements

- **Certifications:** LangChain Academy; Google Cloud, BigQuery ML
- **3rd Place**, RAISE Summit AI Hackathon Paris; **Winner**, Hack for Good (MoodSync); IEEE Xtreme 15.0/16.0; IEEE, ITLAB & AIESEC member
- **Languages:** English, fluent (IELTS) | French, fluent